NUCATALOGUE: WEB-BASED INVENTORY MANAGEMENT SYSTEM FOR NU LAGUNA MERCHANDISE

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Faculty of College of Computer Studies

NU Laguna

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ABSTRACT

The National University of Laguna's Assets Management Office is facing a significant problem in terms of Monitoring of the merchandise stocks and recording of customer data. The current process that the Assets Management Office uses for monitoring stocks is manually counting the stocks through their sizes and types in Excel per table, which extends their time instead of finishing it on time. And for the recording of customers' data, it is manually written in a printed receipt as their acknowledgement for the payment in the accounting office. This study aims to develop an Inventory Management System that keeps Monitoring of stocks and recording of customer data easily, and it has an automated daily sales report to see the total amount received on that day or other day before the present day. This study finds that the new inventory system developed by this study is more accurate and efficient, reducing the processing time and improving the consistency of the asset management. Also, this research study's findings in the area of inventory and stock management will be extremely helpful to both present and future admin and staff. In this way, users will have better expectations of the system, making it simpler for them to stock things and stay informed about the precise availability of each item.

Keywords: Assets Management, Automated, Inventory System

DEDICATION

This study is dedicated to the researchers who come along the way to make it through, by giving the best of their best, working hard, being dedicated to finishing this study, and having a passion for this study.

This study is also dedicated to the families who have been their source of inspiration and gave them the strength to continue when they thought of giving up, who continually provide their moral, spiritual, emotional, and financial support for this study. And friends of the researchers who gave support and strength. To the faculty who believed in and gave guidance to make this study successful.

The researchers dedicate this to God above, who is always there to guide, make the day safe for everyone, and give them strength, the power of mind, skills, and a healthy life.

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Chapter 1
INTRODUCTION

Background of the Study

In the study about manual encoding of the students' data can take a lot of time and work, thus delaying their process and exposing them to different errors (Breitmeyer, 2015). Hence, the researchers have proposed to create a web-based inventory system where they can encode the students' data, who ordered on the dashboard, and the data will be automatically saved in the table tab.

However, the proponent of (Jenkins, 2020) product marketing manager at Oracle, an inventory system will provide work efficiency and information about sales and inventory. Additionally, an inventory system is an effective and user-friendly solution for managing assets across your entire company, from acquisition up to disposal.

For instance, Panasonic's inventory system, Asset Performance Maintenance, is a thorough asset, maintenance job, and spare tracking system that enables users to improve uptime and reduce operating expenses, and inventory. In addition to that, the browser-based PanaCIM Asset Performance Maintenance system has a proven record of accomplishment of effectiveness in several industries, including utilities and power generation, production of heavy machinery, tools and production machines, automobiles, aircraft, and more. Lastly, inventory management Data from other system components, such as inventory control, is governed by aspects of inventory management. Not only that, but inventory management also manages the business operations that take place before the stock enters a warehouse and the routes that the merchandise takes to get to other locations. Features such as multi-location warehouse management tools and integrations with other software such as enterprise resource planning (ERP) platforms (for example, generating

paperless invoices and purchase orders) and understanding the differences between inventory control and management are advantages in a business or enterprise that caters a huge scale of customers.

Currently, the NU Laguna Assets Management Office encodes its inventory and business transactions manually in Excel, a spreadsheet that can perform calculations but is prone to error, making the Asset Management Office's task even more challenging.

Statement of the Problem

The following statements are the current problems that are manifested in the NU Laguna Assets Management Office. The proponents came up with the following:

The Assets Management Office personnel manually monitor the number of stocks of each item or piece of merchandise purchased by customers. In order to solve this problem, we come up with an idea of inventory managing levels which they can track real- time inventory levels. In terms of transactions, some of the customer's data will not be encoded due to a large number of customers, and they will have to manually encode it on a hardcopy sheet. In this problem, we come up to an solution that in our inventory management we will need to input the data of the customer information to track their status so, they won't forget to encode the customers information. the problem of running out of stock, which can result in lost sales, dissatisfied customers, and damage to the brand's reputation. In this problem, we come up with idea of adding information about suppliers and deliveries which they can track and easily order to the supplier and they can track the delivery. And another problem is inadequate monitoring and tracking of inventory levels, which can result in poor decision-making and ineffective resource allocation. In this problem, we come up to an solution of sales reports which they can track the daily sales so, they won't have

the problem for ordering products because they can see the trend sales which indicate the sales patterns followed by the customer.

The main objective of this system is that researchers want to create a sales report system and inventory management system. In such a manner, the users will have better prospects for the system making it easier for them to stock items and be alert about the conciseness of the availability of each item. The specific objectives of this study indicate users can see the pending items when it is out of stock, users are able to view the display report regarding their daily sales, users can be able to add an item when something new arrives, for the system can track the deliveries, efficiently record information about students without a repeating error, create two interfaces with the two users which are admin and staff, and lastly.

Theoretical Framework

This study is based on an economic order quantity model, which refers to the ideal order amount an organization should place to reduce its inventory costs, including holding charges, shortage costs, and ordering costs.

In this part, we know that ordering costs depend on and vary depending on two factors: the expense of ordering too much or too little. These two variables move in opposition to one another. Buying more than you need will increase your inventory costs. Conversely, placing fewer orders will raise the cost of ordering and restocking. Ordering costs are independent of the order size, procurement of items, and stock replenishment.

Storage, labor, shipping, handling, item replacement, shrinkage, and depreciation are all costs associated with keeping products on the shelves at a warehouse and distribution facility. Carrying cost holds the stock, sales rate, incoming deliveries, and suppliers. It is a typical

holding cost, another name for inventory carrying costs.

Replenishment costs are caused by shortages like low stock levels and when customers are not being served because of a lack of stock items. For organizations, effective inventory replenishment has many benefits, such as keeping from running out of stock or overstocking, cutting expenditures like transportation, and increasing customer happiness. Since different items may need to be refilled at various intervals and various rates, the task of inventory replenishment can be incredibly difficult.

Inventory management system is a theory that refers to the process of managing and controlling a company's inventory levels to ensure that the right amount of inventory is available at the right time, in the right place, and at the right cost. The theory encompasses various techniques, such as demand forecasting, inventory optimization, and supply chain management, to help businesses manage their inventory efficiently and effectively.

At its core, the theory of inventory management system seeks to strike a balance between the cost of holding inventory and the cost of running out of inventory. Holding too much inventory can tie up capital and lead to additional storage costs, while having too little inventory can result in lost sales and dissatisfied customers. The theory provides various models, such as the Economic Order Quantity (EOQ) and Just-In-Time (JIT) inventory systems, that help organizations determine the optimal level of inventory to maintain to balance these costs.

Effective inventory management systems can provide numerous benefits for businesses, including improved cash flow, reduced storage costs, increased customer satisfaction, and better supply chain visibility. By implementing a sound inventory management system, businesses can gain a competitive advantage in their respective industries by ensuring that they have the right products available to meet customer demand while minimizing costs.

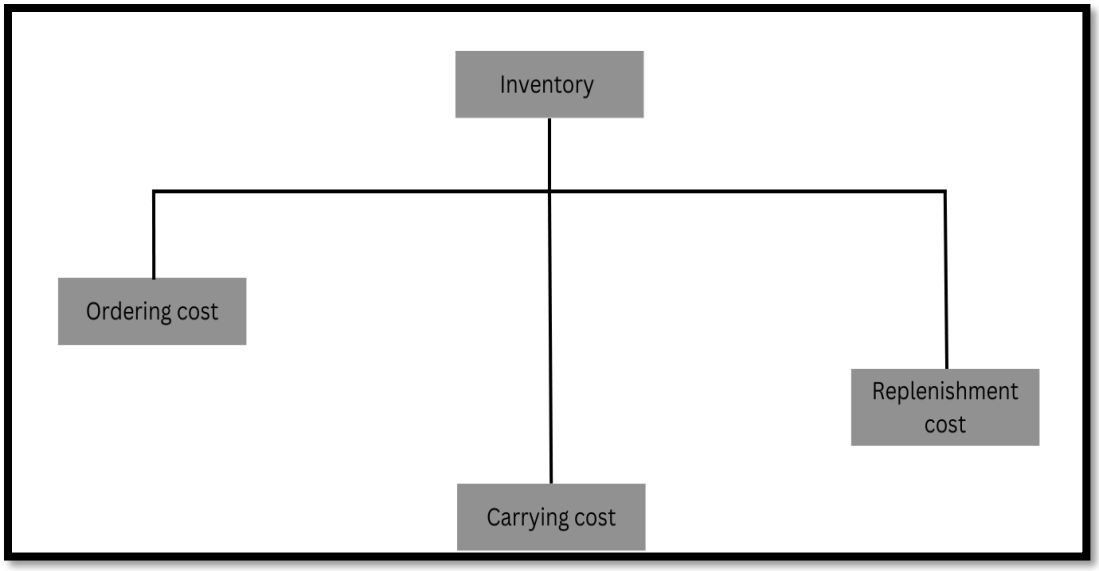

Figure 1. Economic Order Quantity Model

Conceptual Framework

The IPO model (Input Process Output), which was indicated in the theoretical framework, will be the conceptual model applied in this project. The three component which are the input, process, and output—will make up the system. They will act as the constituent parts of the inventory management system, for which this project is entitled NUCATALOGUE.

INPUT

This study provides all the information that is input in our system. There are several features in our input, and this is product information such as name, size, and quantity. In the second feature this is sales and purchase history, in the third feature this is customer information such as name, contact details and status if paid or unclaimed. The fourth feature is supplier information such as name, address and contact details and status if paid or unclaimed. The fifth feature is ordering information such as quantity, date of order, customer/suppliers, and payment status and lastly the deliveries information such as arrival date, quantity, and name. These are the features that are

connected to the IPO model.

PROCESS

In this component process provides all the procedures in the input which are in the first feature product information stored the record of all products information in database including name, size, and quantity. The second feature sales and purchase history processing the track of sales and purchase history with managing inventory levels. Third feature is managing customer information to track sales and provide better customer service. The fourth feature supplier information is to maintain supplier information to keep track of inventory levels and ensure timely delivery of products. The fifth feature is order information which includes purchase orders to order products from suppliers which will track delivery and update inventory and use sales of orders to fulfill customer orders provides better service and lastly deliveries information processing the track of deliveries and delivery history which is arrival time, date, and name of supplier.

OUTPUT

This part provides all the output for every component that has been written in the input and process. In the first output real-time inventory levels provides to keep better track of their inventory levels. The second output this provides sales reports in inventory levels such as purchase history. The third output purchased order to suppliers for restocking products which aligned with customer this explains when customers purchased a product. The fourth output sales orders to fulfill customers' orders which aligned with supplier. The fifth output when the products are out of stock or when inventory levels are low, this will alert or remind the user that they needed to restock products. Lastly the reports on delivery times which provide information on how long it takes for

goods or services to be delivered from the time they are ordered.

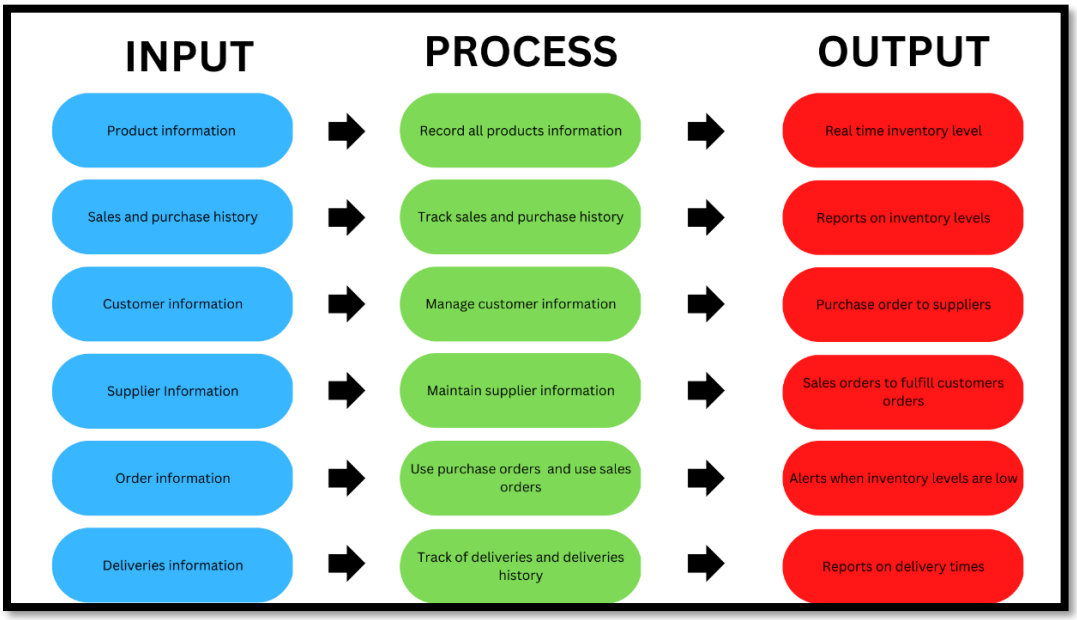

Figure 2. Input Process Output Model

Scope and Limitations of the Study

The researchers created an inventory system that will assist with the tasks of the Assets Management Office at NU Laguna. The primary user of this system will be the staff of the Assets Management Office of NU Laguna and the manager of the Assets Management Office for user access, the system will also remove the difficulty in collecting data on students who buy merchandise like school uniforms or something else under their management except for that equipment.

The NU Laguna's Asset Management Office manually enters students' data into a spreadsheet, checks it for errors, and calculates sales, which takes a lot of time and effort. In the system developed by the researchers, researchers simply will have to encode data such as a student's number, name, and section. This information will be automatically saved in the database.

The system will also keep track of the status of the order, which is noted on the printed

receipt. This system will be available via the web interface, making data submission easier and more convenient. The inventory management system's minimum requirements are Intel i5, 8 GB of RAM, and Windows 10. The researchers interviewed the client to gather information about the client's challenges and to come up with a solution.

Significance of the Study

This study speaks about the study's impact on a research field and its contribution(s). The importance also identifies who and how the research findings are beneficial.

Creating a system that would assist and improve inventory management and sales reports is notably significant as business transactions undergo technological improvement and development. First, the success of this research study will develop and improve the inventory management and sales reports of the Assets Management Office of NU Laguna. This system will accelerate data encoding and computations and provide more systematic methods of keeping information and business transactions.

Second, this study will assist the Assets Management Office of NU Laguna's employees since the system will reduce the process of manually encoding the customer's data and stock data during a transaction. Additionally, this research study will be of major help to current and future business owners and employees in the field of inventory and stock management.

Third, the importance of this study will alter the needs of the employees and customers. This study will make it possible to record data of each business transaction most efficiently and rapidly.

Lastly, this study will supply additional knowledge for future researchers who aspire to explore and innovate current studies about inventory and stock management.

Promotion of value and social relevance. The NUCATALOGUE: WEB-BASED INVENTORY MANAGEMENT SYSTEM FOR NU LAGUNA MERCHANDISE aims to provide an effortless way for the asset management office to record data and monitor stocks using this system. The researchers guarantee that it will be easier for them to inventory items. First, the user will speed up and save time on their monitoring of items, and it will make it easier for them to use it; there is no need to open Excel and create a formula to calculate how many stocks are available. Second, they do not need to count the items that are only available because, in our system, they monitor when an item is in stock if it is only a few or if it is no longer available, for example, a large polo school uniform. It can see the status of this item if there are more or if it is no longer available. The NUCATALOGUE: WEB-BASED INVENTORY MANAGEMENT SYSTEM supplies a lot of help for asset management that handles inventory items, especially in large universities or companies. This inventory system will also be valuable for item registration and monitoring, especially since most of the schools are having face-to-face meetings and there are a lot of students who want to buy a uniform. Using this system, they will be able to speed up what they are doing, and they will not waste too much time making a formula in Excel to calculate the items they have.

Contribution to nation-building. The researchers contribute to nation-building by improving the efficiency and accuracy of inventory management by providing a platform for better tracking of sales and inventory. As aspiring programmers, the researchers reduce waste and improve customer satisfaction, enable businesses to make informed decisions and respond to changes in the market more effectively, and lastly, the researchers contribute to the development of the local workforce and economy.

Contribution to an existing body of knowledge. The contribution of this system in the

field of monitoring stocks and recording data is that it will serve as an inspiration or model for other universities so that they also create an inventory system that will also help them if they also have asset management. This project informs each process of monitoring and recording, which will serve as a solution to that problem. And it also contributes to the body of knowledge because it also enhances the features that are not commonly found in other inventory systems.

Continuous improvement of the teaching-learning process. By use of our web- based inventory management system incorporating technology into the management of materials, supplies, deliveries, sales, and purchased items. The Assets Management Office can focus more on its core responsibilities, such as procuring, encoding, distributing materials, and supplies, and helping to ensure that the necessary materials are available in a timely and efficient manner.

Definition of Terms

The following terms give an understanding of the concepts or factors that will be discussed throughout the study and are used throughout the capstone project and are defined as follows:

Inventory Management System. It monitors the amount of physical inventory you have in your company.

Web-Based System. It provides computer and internet-based access to a software system.

Acronyms

Some commonly used acronyms in the field of technology include:

IMS	**Inventory Management System**
WBS	**Web-Based System**

Chapter 2
REVIEW OF RELATED LITERATURE AND STUDIES

This chapter summarizes the relevant studies and literature from many scholars that support the inventory management system. It also emphasizes the significance of the studies in the proposed system, which are summarized.

According to Bonney (1994), due to international competition and advanced technology, inventory management is changing rapidly, and has had a successful development in years. The inventory system is a procedure that tracks inventory, purchases, and sales along the entire supply chain. Businesses use inventory systems to make sure they are aware of exactly what products are on hand and where they are located. To assist businesses in organizing and keeping track of their inventory, inventory management systems give accurate records of new and returning goods as they enter or leave the warehouse. Additionally, these systems can keep track of information like the quantity sold, the cost per unit, the serial number, the lot number, the dates of purchase, and manufacturing.

Automated Sales Report

Bisagas, Arivada, and Tanteo (2009), manual labor is used to count the rice sacks, which makes it challenging to calculate the number of unsold goods and available stockpiles. One of their fundamental issues ever since the company was founded has been the manual compilation of inventory reports, such as daily, weekly, quarterly, and annual inventory lists of data. The owner's inventory management challenges will be reduced to a minimum by the transaction processing system for Best Choice Rice Dealer. To avoid understocking, overstocking, or stockouts, it will keep an eye on the products' availability for replenishment. The system will also produce the sales

reports and other inventory information that are required. Furthermore, Villegas (2009), stated Computers can only do so much, but in conjunction with encouraging staff members to be honest and putting in place the right deterrents, they can perform extremely effectively in reducing losses and thefts.

In a retail and wholesale business, it is therefore highly recommended to implement both manual and automated inventory systems.

The study of De Leon and Ferrer (2009) developed an automated sales and inventory system known as the Korean Red Ginseng Enterprise Sales and Inventory System. The advocates employed databases to facilitate file access and expedite the selling and inventory transaction processes. The software was made to produce reports like monthly reports, inventory reports, sales invoices, and itemized lists. Additionally, a study from Alday, Espino & Ragudo (2010), entitled "Computerized Sales and Inventory system for Ronmon Trading" was able to show a significant result upon replacing the manual system with their proposed system, in which transactions were processed more quickly and accurately. The aforementioned technology is quite trustworthy; it removed mistakes and erroneous data and produced dependable reports that were beneficial.

Lastly, a study claims that businesses nowadays benefit from embracing current technology to elevate their standing and ensure a more effective manner to make their task simpler (Laranang, Maaño & Nañola 2009). Also, according to this study, the key issue with Osaka Iridology is the need for a modern, efficient instrument to be used in constructing a sales and inventory system that will effectively manage their transaction. So, they create an all-encompassing system for Osaka Iridology's transactions, including tracking of their medication and daily sales.

Reduce of Time

Inventory Management is a vital part of a business for it is the major asset of a company and thus determines the flexibility of a company (Bonney,1994). Additionally, Robert Spector, a retail historian, a company should have a good inventory system for it enables the business owner to forecast demands with much accuracy (Eckert, 2007). The statements above suggest the importance of an accurate and efficient inventory management system. Thus, creating an advanced inventory management system for a certain business is significant because it will reduce the devoted effort and time for asset management, yet personnel can still be supplied with the accurate information they need to perform their responsibilities (Bronack, 2012). Currently, there are different case studies on how an inventory management system is effective when it comes to customer satisfaction and work efficiency.

In a study conducted by Eckert (2007), he quoted Edward Marien, Director of Supply Chain Management at the University of Wisconsin, the definition of a "perfect order," which is when a customer finds the right product, destination, condition, documentation, and cost. This study shows how the use of an inventory system management can produce a "perfect order" which then can affect the customer satisfaction level. The findings of the study showed how the use of a new inventory system management changed the perspective of personnel that was used to the manual encoding of inventory data. Since the use of this inventory system management enabled the distributor to accurately predict demand, business owners reduce loss and satisfy customers better.

Record Keeping

Another study about emerging inventory management techniques stated that there is 12 broad viable inventory control approaches that management students, business owners, inventory

managers, store supervisors, and analysts can benefit from. The context that the study was undertaken by Aro-Gordon & Gupte (2016), Software Application and Tracking System is one of the twelve emerging techniques which survey shows using this approach is a valuable tool for organizations seeking stock control system upgrade. Additionally, customizing inventory management software can also be done to fit your individual needs. Also, this software provides complete inventory control allowing inventory managers to do proper inventory record-keeping, material requirement planning (MRP) of getting the right item to the right place at the right time, and inventory decision- making processes such as anticipating workloads. However, the findings of the study also state that the use of emerging inventory management techniques varies according to the size of the organization, cost efficiency, and space availability.

Synthesis

From the above statements from different foreign and local studies, the significance of having an inventory system management software was emphasized specifically from the business owners, personnel, and customers' perspective. It also emphasized the efficiency of this kind of software in the field of stock control management.

In the study of Eckert (2007) and Alday, Espino & Ragudo (2010), positive results of using an inventory system management instead of the manual process were observed by both the personnel and customer with quick and more accurate data. Finally, some studies show that creating software that can efficiently control stock and manage business transactions like record-keeping, automated computation, making sales report, etc. are the needs of the current organization or company as the competition and advancement in technology arise (Laranang, Maaño & Nañola 2009, De Leon & M. Ferrer, 2009 and Aro- Gordon & Gupte, 2016).

Chapter 3
METHODOLOGY

This chapter presents the project design, project development, operation, and testing procedure as well as the evaluation procedure.

Project Design

The Web-Based Inventory Management System was developed to help the NU Laguna Assets Management Office facilitate their work. The system can save data of students who order merchandise like Traditional School Uniform, PE Pants & T-Shirt, NSTP Shirt, etc., and monitor the number of stocks of the Items if they are low or there are more.

The functional decomposition diagram shows the role of the administration. In our web-based inventory management system, the only users are the staff and manager of the asset management office, so students will walk in and get the receipt. Once they get the receipt. They need to look for their sizes. Once they get their sizes they will go to the cashier and pay. After that, they will go back to the assets and the admin will give the uniform, or if the uniform is out of stock, they will be informed when it can be claimed. And also, the admin can see if the customer status is claimed or pending. The Staff will print receipts to give to students who order merchandise before they proceed to the accounting office for the payment. This only shows the task of the admin and Staff in our web-based Inventory Management System.

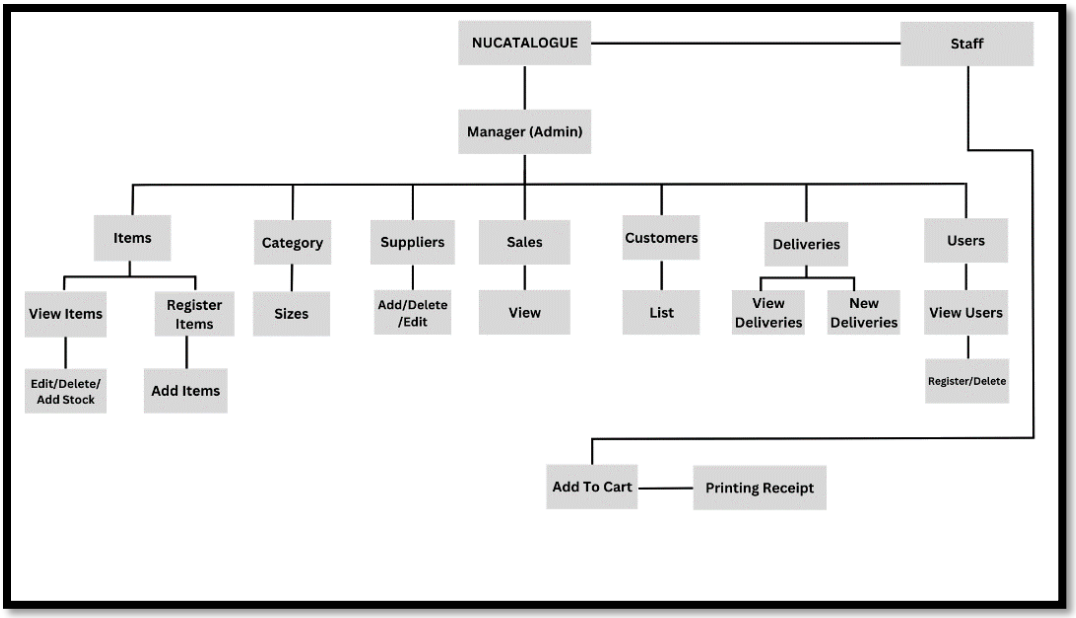

Figure 3. Functional Decomposition Diagram

Project Development

The Iterative Life Cycle Model was used by the researcher in developing the Web- Based Inventory Management System. This model can go back to the previous phase to facilitate improvement.

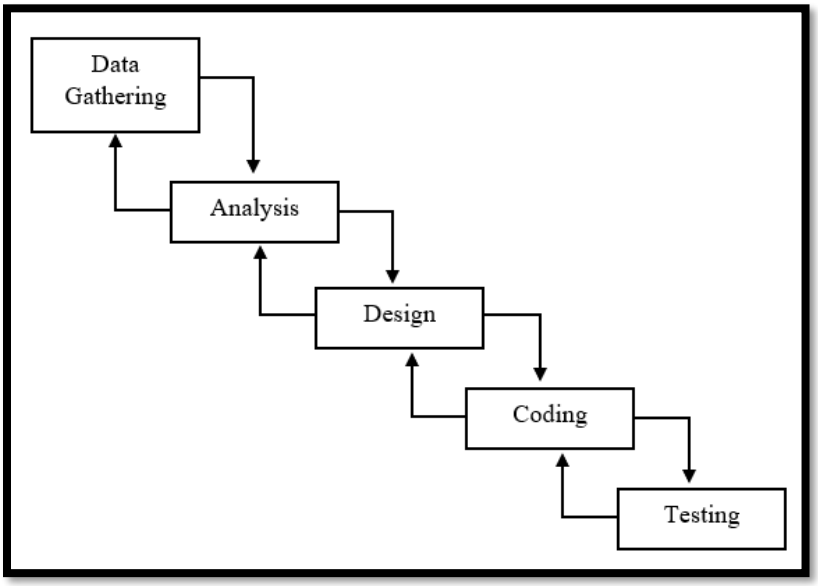

Figure 4. Iterative Life Cycle

Data Gathering

For the data collection method, researchers gathered systems they were familiar with and filtered down the possible clients that they would take. After narrowing their strengths and weaknesses, the researchers approached a client that would present obstacles but was still manageable. The researchers approached the target client to conduct an interview and understand their issues and desired system.

Analysis

The current encoding process of the Asset Management System is the manual encoding of data in excel spreadsheet format. In this process, the name of the student, order receipt number, date, quantity of orders, size, and type of uniform are manually encoded thus taking a lot of time and effort and prone to human error. For these reasons, the target client requested automated system software that can efficiently do these manual tasks. Furthermore, the researchers added some more features that can make the clients work more efficiently.

DATE	NAME	OR #	P.E SHIRT	QTY	P.E PANTS	QTY	NSTP SHIRT	QTY	VARSITY JACKET	QTY	POLO	QTY (XS-L)	QTY (XL-6XL)
June 6, 2022	[illegible]	[illegible]											
	[illegible]	[illegible]									M	1	
June 7, 2022	[illegible]	[illegible]											
June 8, 2022	[illegible]	[illegible]	M	1	L	1							
	[illegible]	[illegible]							2XL	1			
June 10, 2022	[illegible]	[illegible]											
June 13, 2022	[illegible]	[illegible]	XL	1	XL	1					XL		2
June 15, 2022	[illegible]	[illegible]	3XL	1	3XL	1			2XL	1			
	[illegible]	[illegible]							XL	1			
June 17, 2022	[illegible]	[illegible]	M	1	L	1							
June 20, 2022	[illegible]	[illegible]					M	1					
	[illegible]	[illegible]									XL		1
June 21, 2022	[illegible]	[illegible]					L	1	L	1	M	1	
	[illegible]	[illegible]					S	1	M	1			
June 22, 2022	[illegible]	[illegible]											
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June 23, 2022	[illegible]	[illegible]											
	[illegible]	[illegible]											
June 24, 2022	[illegible]	[illegible]							M	1	M	1	
June 27, 2022	[illegible]	[illegible]											
	[illegible]	[illegible]									M	2	
June 28, 2022	[illegible]	[illegible]							M	1			
	[illegible]	[illegible]											
June 29, 2022	[illegible]	[illegible]					XS	1					
	[illegible]	[illegible]											

Figure 5. Current Encoding Process of Asset Management System

Design

In terms of design, the context diagram and data flow diagram of the system is shown here. The usability or benefit of the system was assessed and verified by the researchers in this phase.

Context Diagram

The context diagram in this section depicts the two roles, which are manager and staff. In essence, Admin will have the power to manage the inventory, including the ability to add, update, and delete things. The admin, on the other hand, can add, amend, or remove user accounts and give them to the employees. The admin can also add, remove, or update suppliers and incoming deliveries so they can keep track of when and at what time the delivery was received. Also, staff members can contact suppliers and let them know they have an out-of-stock item. And also, the staff can print a receipt for the payment.

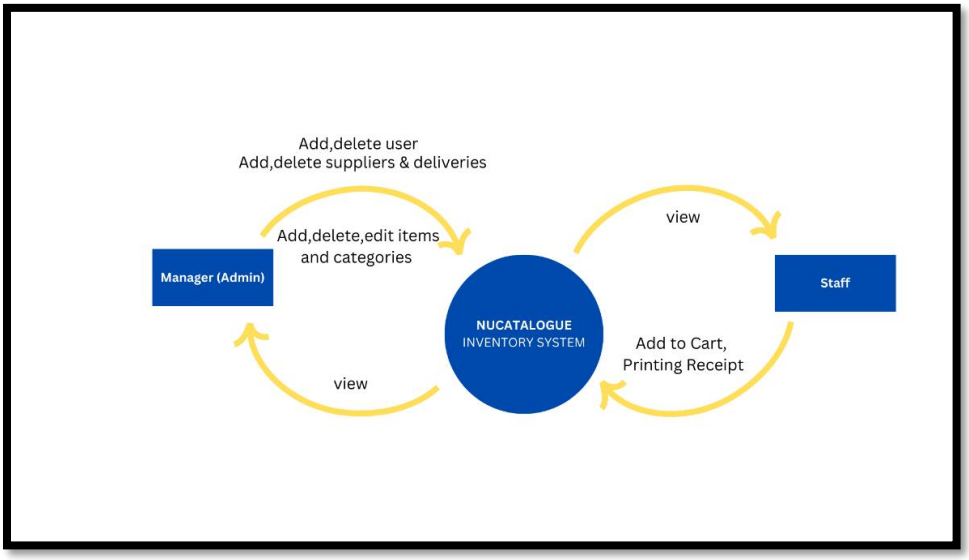


Figure 6. Context Diagram

Data Flow Diagram

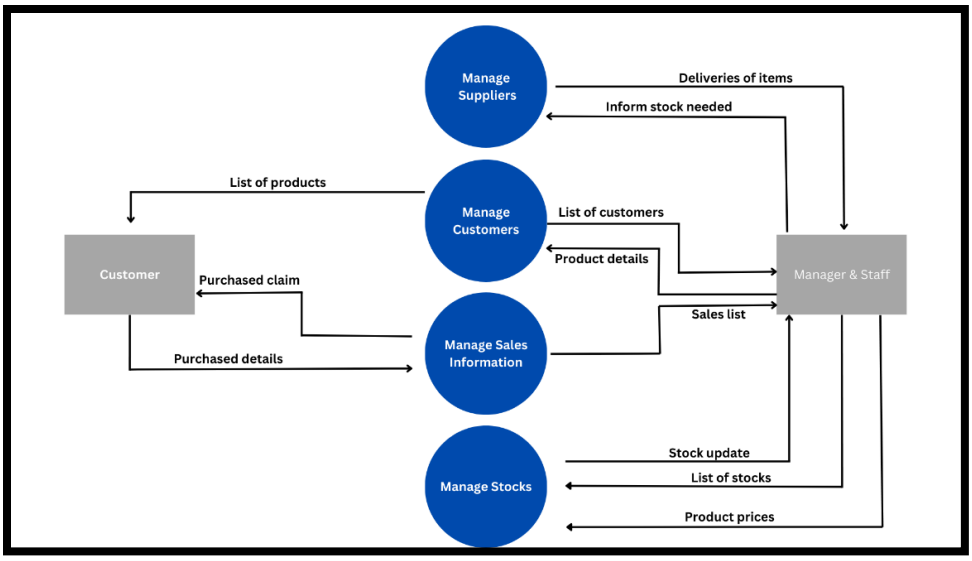


Figure 7. Data Flow Diagram

In this operation of our system of data flow, customers and staff will be external entities. It displays all the information that is transferred throughout each step, such as the duties they have as clients or employees. We utilized words like suppliers, customers, sales, and stockpiles to describe the process. As a result, it displays the input and output of every data flow in an external entity that reaches employees and clients, which displays fewer data flows as they will just need to buy an item like uniforms, etc. This project, which is entitled NUCATALOGUE, will be web-based, and by saving all the data of the items and products, we used CRUD operations with PHP and MySQL. The customers will walk into the NU Laguna Assets Office so they can get the list of products. On the other hand, staff will give them attention and process the item that they order. After that, they need to go to the accounting department for the payment fee for the item. Once they receive the receipt, they can claim the item, and staff will encoding the purchased item in detail.

Coding

As for Coding, the researchers proceeded after they finalized and implemented the design. Here in this part, the researchers used coding software Visual Studio Code and Notepad for testing, and they also used PHP and SQL languages because the system has a database. So, this is what the researcher's used software because the system is Web-Based.

Testing

After the coding, the researchers conducted the testing where they will evaluate if the system works or if there is anything that needs to be changed or improved. The researchers tested the system using Xampp Server and localhost for the Database. Since the system is new, it is expected that it should have a bug, but it is easy to fix it because the researchers conducted testing and when the error is identified, they will immediately fix it.

The system works well on the PC with the following specifications:

- AMD Ryzen 5 3600 6-Core Processor 3.59 GHz
- RAM 16.0GB
- Windows 8/9/10/11

The system works well with the laptop, with the following specifications:

- Display size (inches) 15.6 inch.
- Windows 10/11
- AMD Ryzen 5 3500X Processor
- RAM 8GB

This method involved evaluating the software to identify any flaws and necessary improvements. When errors were discovered, the researcher changed the system right away.

Operation & Testing Procedure

This section identifies the different operation and testing procedures of the developed system to ensure the completeness and accuracy of the developed application software.

Software Access Procedure:

1. Download the ZIP file.
2. Download Xampp 7.3.0 and install.
3. Extract the ZIP file into htdocs inside the xampp folder.
4. Run xampp control panel and open mysql.
5. Input localhost/NUcatalogue in URL box.

System Operation Procedure

1. Featured Items can be viewed and register, as for view you can edit/delete/add stock, and for the register, you can add items/merchandise.
2. A category is where you can input sizes.
3. Suppliers where can you input the information of your suppliers and you can edit/delete it.
4. Sales are where you can track your sales as viewed as a table or viewed as a graph. Then you can see the time and date of the purchased items.
5. Customers are where you can view the list of your customers' information. Tracking of deliveries is where you can view the details of deliveries.
6. The users' function is where you can view the list, you can register/delete.

System Testing Procedure

1. The developed web-based inventory management system will be installed on our client's laptop or client's desktop. The chosen guest is our client who will evaluate and evaluate the software.
2. Users were given the default username and password for our client who is the admin.

3. When the admin already logged in; features were presented as the following features for admin users.

   a) **Items Feature**

      - View items can edit/delete/Add stock.
      - Register items where you can add the latest items.

   b) **Category Feature**

      - Add size

   c) **Supplier Feature**

      - Add supplier.
      - Delete supplier.
      - Edit the supplier if you have added the supplier.

   d) **Sales Feature**

      - View as graphical.
      - View as a table then view the list.
      - View the last 7 days sales.
      - View the monthly sales.
      - View the yearly sales.

   e) **Customer Feature**

      - View the customer list.

   f) **Deliveries Feature**

      - View Deliveries
      - New Deliveries input the needed information if you have delivered stock.

   g) **Users Feature**

      - Register new user.
      - Delete user.
      - View list user

Evaluation Procedure

NUCatalogue was implemented on a computer for testing. After that, this study was conducted at NU Laguna by interviewing the client and observing the problem they were facing. During the testing, a rating scale with several 1s to 5s per criterion will be used as the grade in the evaluation. The tools used by the researchers in the evaluation are the interview and observation while conducting the testing, and their respondents are the staff of the NU Laguna Assets Management Office because they will also benefit from this if this system and study become successful. And our respondents will give a rating based on the evaluation conducted.

Mean scores were used to interpret the results of the evaluation conducted. However, the Likert Scale below was used as the basis for the interpretation of the data.

Table 1

Likert Scale for the Interpretation of Data

Numerical Scale	Descriptive Scale
4.51 – 5.00	Excellent
3.51 – 4.50	Very Good
2.51 – 3.50	Good
1.51 – 2.50	Fair
1.00 – 1.50	Poor

Chapter 4
RESULTS AND DISCUSSION

This chapter presents the project description, project structure, project capabilities and limitations, and project evaluation of the study.

Project Description

The researchers developed an Inventory Management System for NU Laguna Merchandise. The system software is accessible by using any of the two users (manager or administrator and staff) which have their own access rights and levels.

The system software made by the researchers can perform the following:

1. Monitoring of Stocks
2. Record Keeping
3. Automated Daily Sales Report

Monitoring of Stocks is simply done by viewing and registering items when there are new deliveries or new arrivals. Also, the system will keep records of customers who ordered merchandise, so that in case of paid merchandise were unclaimed, the staff or the administrator can contact the customer. Additionally, the system is programmed to know when the user needs to restock once the available quantity of merchandise reaches 50 and below until it reaches out of stock. Lastly, the system can provide and display a daily sales report.

To run the system, a desktop with the specifications of AMD Ryzen 5 3600 6-Core Processor 3.59 GHz, RAM 16.0GB, Windows 8/9/10/11 and a laptop with the specifications of Display size (inches) 15.6 inch, Windows 10/11, AMD Ryzen 5 3500X Processor, RAM 8GB are recommended by the researcher for easy and smooth use of the system.

Project Structure

The system software was programmed to be user-friendly and appealing to the eyes of the user thus, the researchers created a home page for the software where the user can easily log in to the main page of the software with the use of their username and password. The researchers made a homepage that is personalized and pleasing to the eyes of the user.

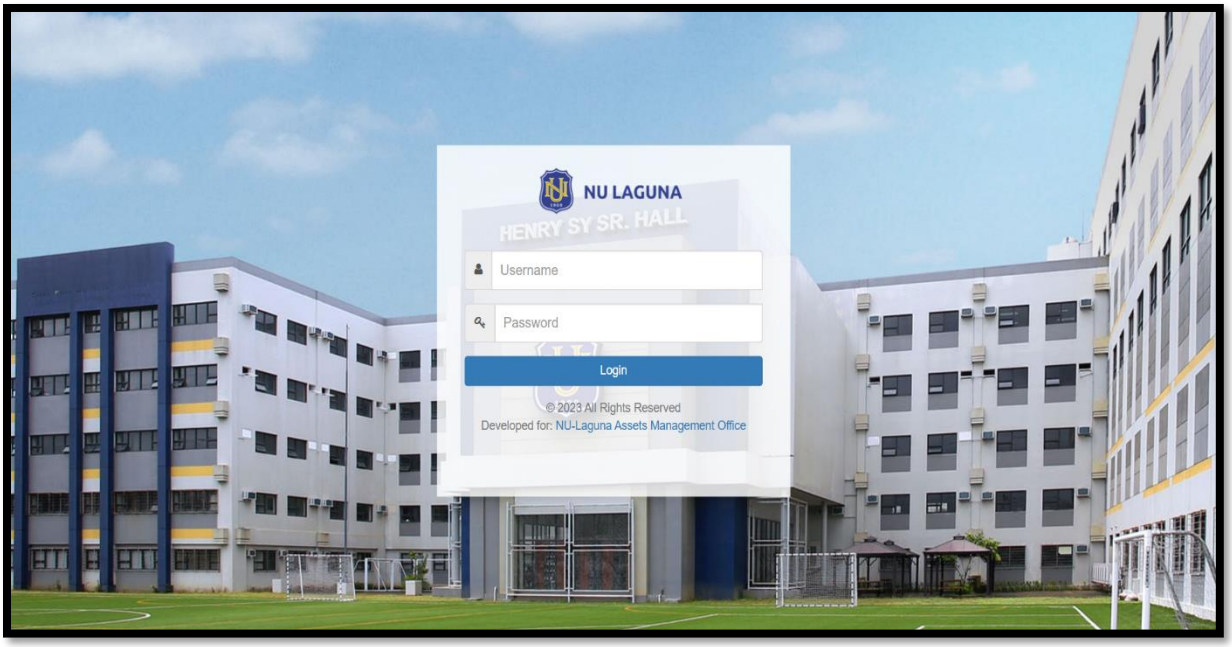


Figure 8. Log In Page for Admin

For the main page of the software, the researchers programmed an automated inventory management system in which users can efficiently add, edit, delete, and restock items. Also, upon accessing the main page the user can view and monitor the status of the items.

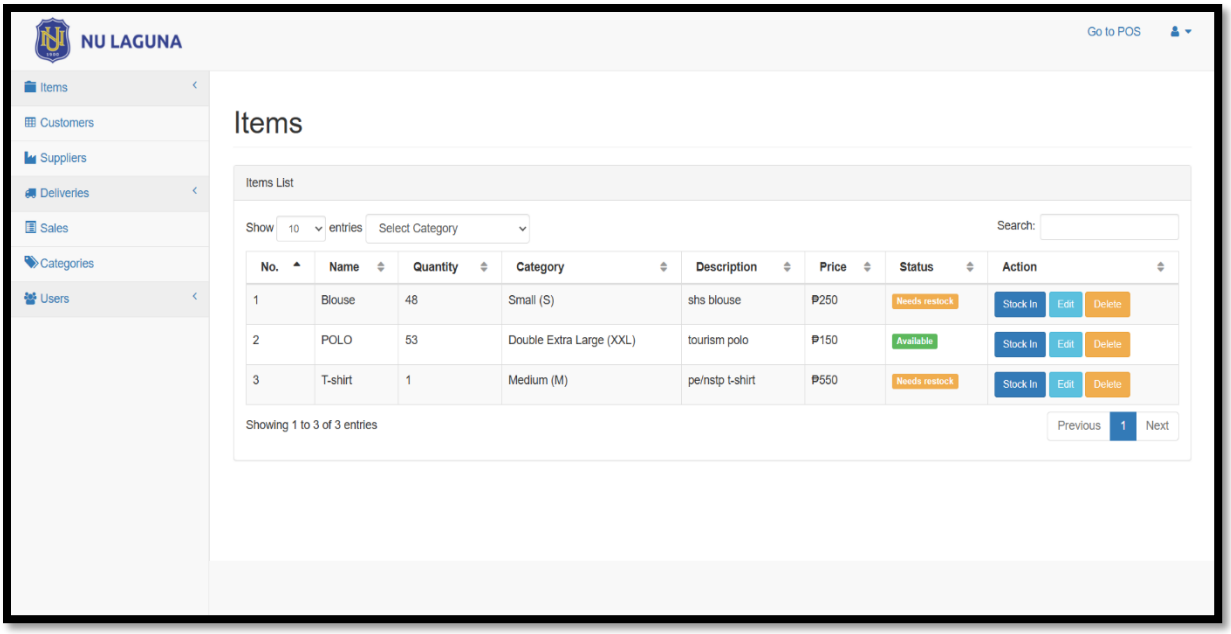


Figure 9. View of Items

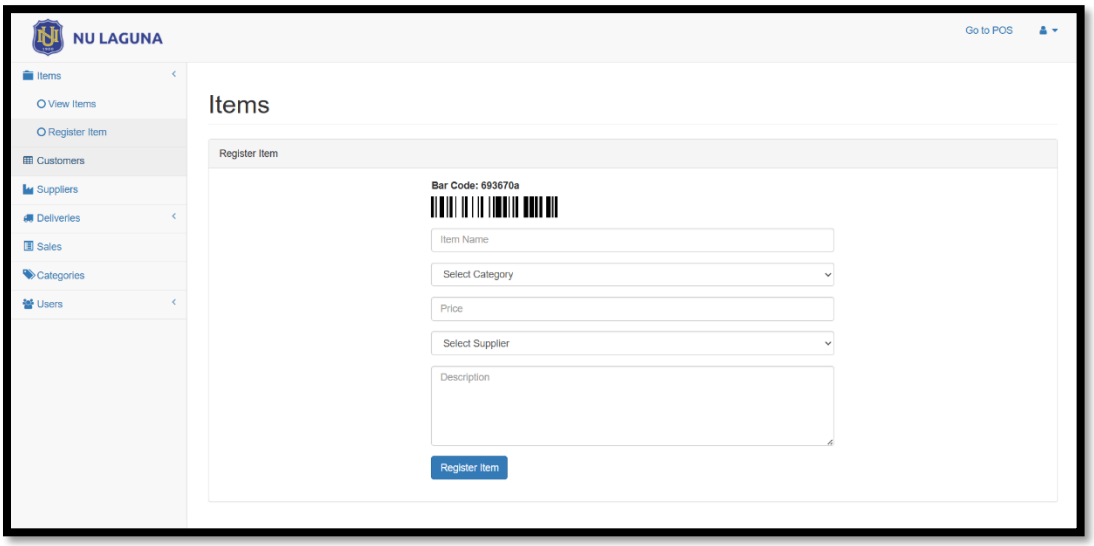


Figure 10. Adding New Items

The software system was also designed to keep the customer's pieces of information.

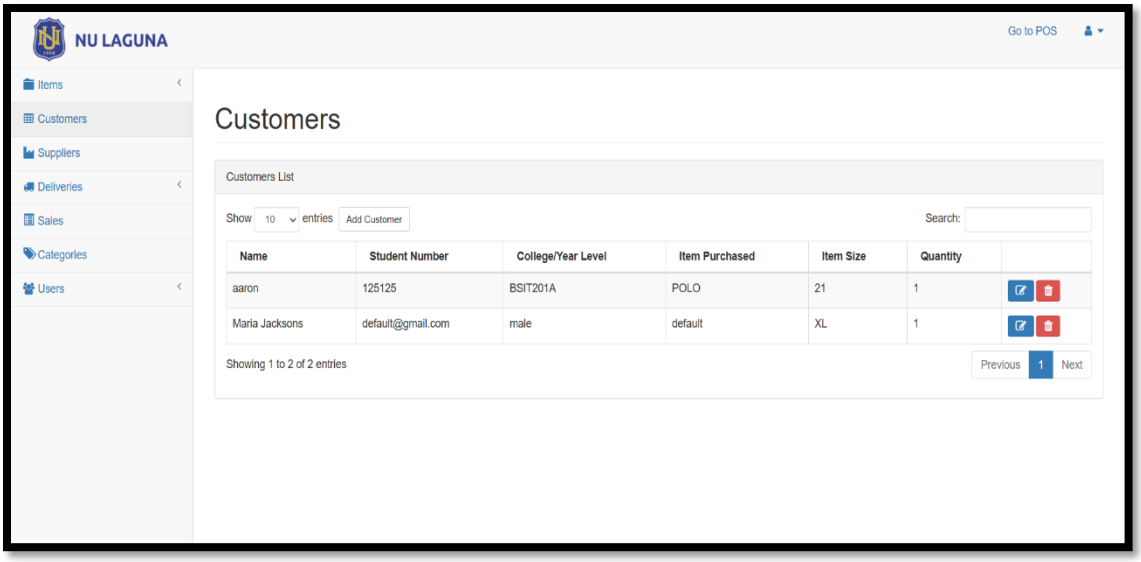


Figure 11. Customers List

The researchers also designed a page where information about deliveries such as date and time of arrival, suppliers' names, items, and quantity. The platform will include a user interface that allows users to easily add and view the information. This will make it easier for users to find the information they need.

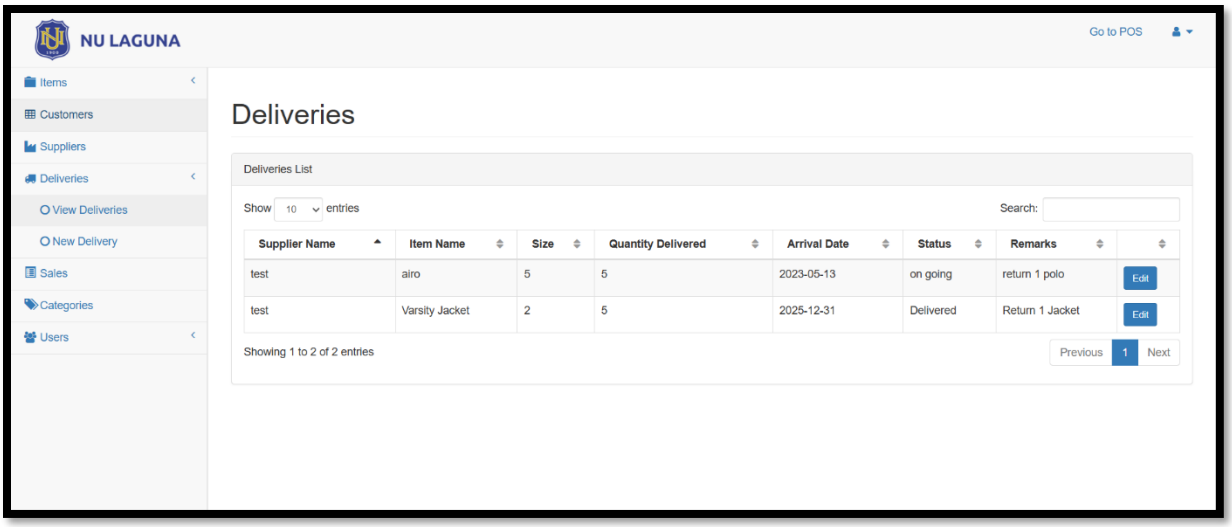
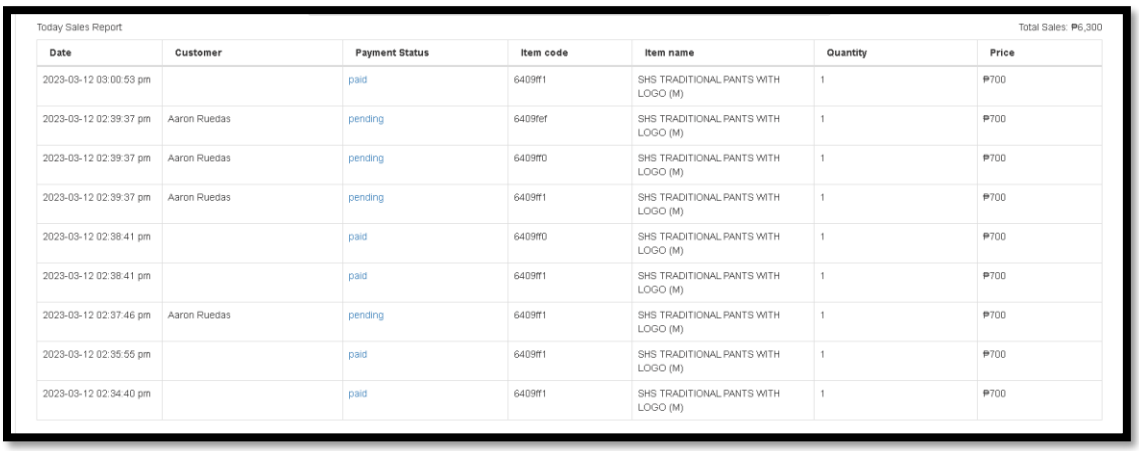


Figure 12. Expected Deliveries List

The software system was also programmed to display sales reports that can be viewed in table form and graphs. This page also displays the payment status of a customer. Thus, this page is focused on providing users with accurate information and sales reports. Lastly, the user interface is designed in a way that is intuitive and user-friendly, with clear visually appealing design elements.



Today Sales Report

Total Sales: ₱6,300

Date	Customer	Payment Status	Item code	Item name	Quantity	Price
2023-03-12 03:00:53 pm		paid	6409ff1	SHS TRADITIONAL PANTS WITH LOGO (M)	1	₱700
2023-03-12 02:39:37 pm	Aaron Ruedas	pending	6409fef	SHS TRADITIONAL PANTS WITH LOGO (M)	1	₱700
2023-03-12 02:39:37 pm	Aaron Ruedas	pending	6409ff0	SHS TRADITIONAL PANTS WITH LOGO (M)	1	₱700
2023-03-12 02:39:37 pm	Aaron Ruedas	pending	6409ff1	SHS TRADITIONAL PANTS WITH LOGO (M)	1	₱700
2023-03-12 02:38:41 pm		paid	6409ff0	SHS TRADITIONAL PANTS WITH LOGO (M)	1	₱700
2023-03-12 02:38:41 pm		paid	6409ff1	SHS TRADITIONAL PANTS WITH LOGO (M)	1	₱700
2023-03-12 02:37:46 pm	Aaron Ruedas	pending	6409ff1	SHS TRADITIONAL PANTS WITH LOGO (M)	1	₱700
2023-03-12 02:35:55 pm		paid	6409ff1	SHS TRADITIONAL PANTS WITH LOGO (M)	1	₱700
2023-03-12 02:34:40 pm		paid	6409ff1	SHS TRADITIONAL PANTS WITH LOGO (M)	1	₱700

Figure 13. Sales Report

This structure provides the users (staff) with information concerning the availability of items for the customer. After the customer provides information about his/her needed item and quantity the user can input it into the system.

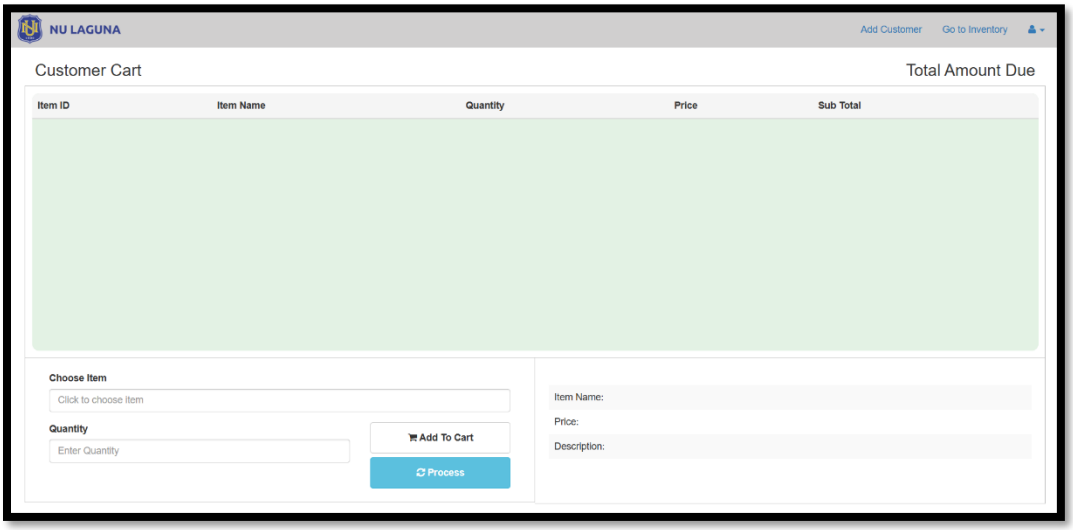


Figure 16. Customers Cart

This structure shows the procedure after you confirm the items of the customer. As for the data of the customer it will be saved in the sales report feature where you can view the status of the customer if he/she is paid or unpaid. All these components were developed to give users a seamless and visually pleasing experience.

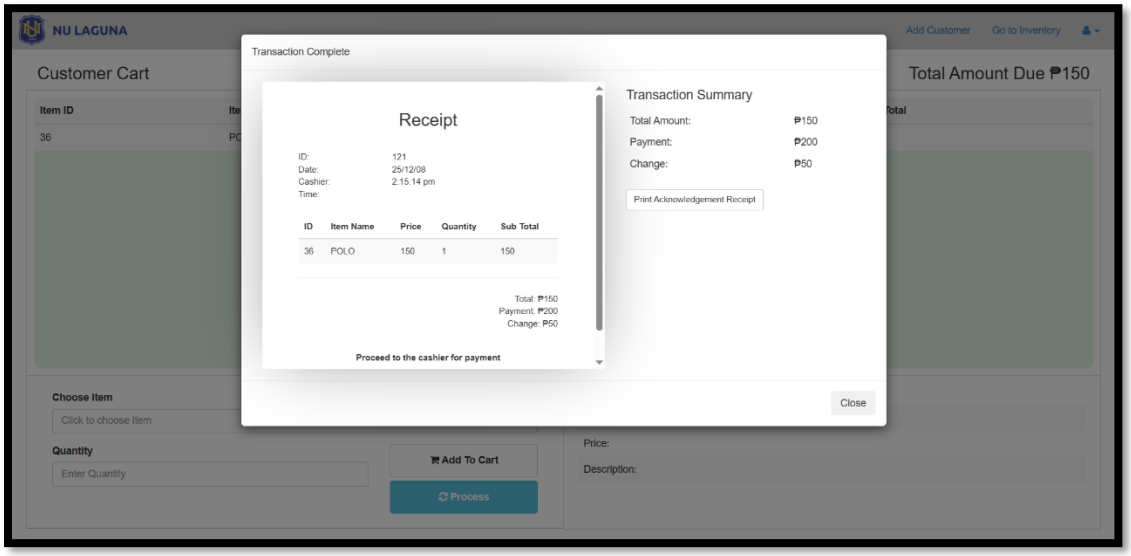


Figure 17. Printing Acknowledgement Receipt

Project Capabilities and Limitations

The system has the following capabilities:

- The system can display the quantity of the items.
- The system can automatically change the status of each item based on quantity.
- The system allows the user to view the status of the customers if claimed or unclaimed.
- The system can generate the acknowledgment receipt.
- The system can display the sales history of the items.
- The system can track the arrival time.

The developed system has limitations that are workable and may still be improved, such as the following:

- Scanning of barcode
- Delivery Tracking
- Item tracking

Project Evaluation

The following criteria were used to evaluate the produced system: (a) Functionality; (b) Usability; (c) Reliability; (d) Efficiency; and (e) Maintainability. Staff members who will utilize the system during its pilot period evaluated these criteria. Personnel from NU Laguna's Assets Management Office, the testing facility, are included in the respondent. The mean evaluation ratings for each criterion are displayed in Table 3.

Table 2

Overall Evaluation Results

Criteria	Overall Mean Score	Descriptive Rating
a. Functionality	4.33	Very Good
b. Usability	3.66	Very Good
c. Reliability	3.33	Good
d. Efficiency	4.00	Very Good
e. Maintainability	3.66	Very Good
Overall Mean Score	3.79	Very Good

The project scored highest for functionality and lowest for reliability, according to the evaluations. The term "functionality" describes a system's capacity to carry out its planned

functions and adhere to its specifications (citation). The system's functionality in this area obtained the highest score, indicating that it fulfilled the testers' expectations and was satisfactory. Usability is a term that describes how simple and user-friendly a system is to use. The system was given a middling rating in this category, meaning that it was generally simple to use but could be made more user-friendly. Any component of a computer system (such as software, hardware, or a network, for instance) that regularly functions following its requirements is deemed reliable. This area received the lowest rating in the evaluation. This might be brought on by problems with the system's speed, stability, and capacity to manage rising demand.

Chapter 5
SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the summary of findings, conclusions, and recommendations based on the results of the evaluation, comments, and suggestions.

Summary of Findings

The developed Inventory Management System can monitor the number of stocks and can perform a restock action in case the number of any item runs out. The system can also save records of customers who buy Merchandise as a contact to provide information and ease of access if the customer has claimed the merchandise. Based on the results of the study, the following are the summary of the findings:

- **Functionality**. The respondents rated the system very good because it helps the user to view the records of who bought the items, track the current stocks of the item for each category, and can only be accessed by authorized users.
- **Usability**. The respondents rated the system very good because it uses words that can be easily understood and has buttons that indicate the functions. And it can be easily learned by all the users due to its user-friendly interface, which meets their expectations.
- **Reliability**. The respondents rated the system good because from time to time the other features show errors and the system is still in the process of troubleshooting.

Conclusions

In consideration of the objectives of the study and the results of the evaluation, the

following conclusions were drawn:

1. The development of NUCATALOGUE: A WEB-BASED INVENTORY MANAGEMENT SYSTEM FOR NU LAGUNA MERCHANDISE has provided an automated way to monitor item stocks.
2. The system is designed to meet the needs of the Asset Management Office to provide additional assistance in regard to the client's difficulties with manually checking the items and monitoring each item; thus, providing efficiency in their work for the system will provide the records of the items that they have.
3. The system may have the potential to be useful for the client, especially in the Assets Management Office, based on the result of the test conducted, which rated in category of Functionality and Usability as "Very Good" except in Reliability as "Good".

Recommendations

For the improvement of the study entitled NUCatalogue, it is recommended:

1. The system should create a new function that allows workers to easily add items by scanning a bar code that will input the item's name and price on the system.
2. The system should enhance the sales tracking, which will display the most items bought.
3. The delivered items should be tracked by the system, and automatically add delivered items into the system.

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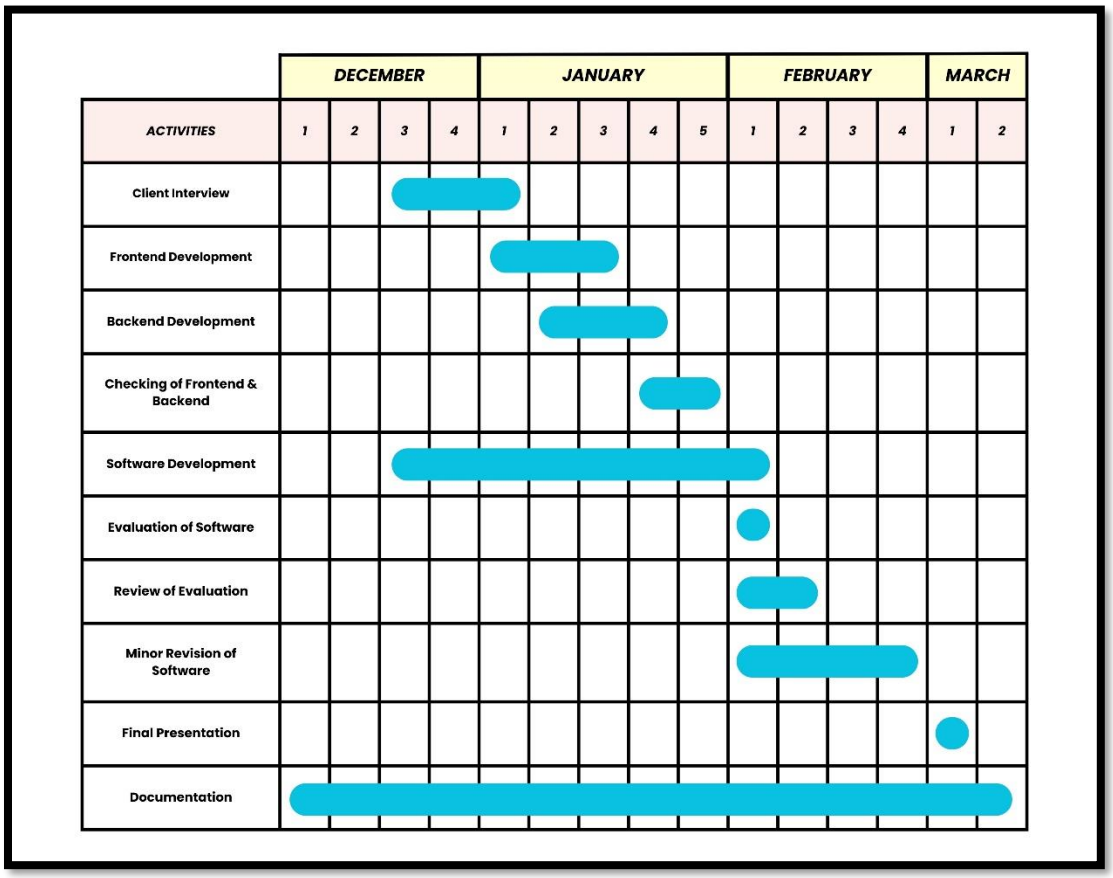
APPENDICES

Appendix A
Gantt Chart

NUCatalogue Development Activity



Appendix B

Copy of Research Instrument

Sample Evaluation Criteria for Software Materials

Instruction to the Evaluator: Please evaluate the prototype by using the given scale and placing a checkmark (√) under the corresponding numerical rating:

Table 3

Category

NUMERICAL RATING	EQUIVALENT
5	Excellent
4	Very Good
3	Good
2	Fair
1	Poor

Table 4

Evaluation Questionnaire

INDICATORS	5	4	3	2	1
A. Functionality **1.** The inventory system allows for easy tracking and monitoring of inventory levels, reorder points, and lead times. **2.** The user can easily add, edit, and delete items. **3.** The system can provide an accurate daily sales report. **4.** The system can help the user track and add new arrival deliveries.					

5. The system can aid the user in viewing the status of each item. **B. Usability** **1.** The ease of navigating the inventory system is a factor to consider. **2.** The system functions can be easily understood. **3.** The system's design is user-friendly. **C. Reliability** **1.** The inventory system provides accurate and real-time information on stock levels, inventory movements, and product availability. **2.** The system keeps the information of customers, suppliers, and products secure and private.					

Table 5

Summary

Summary	**Sum**
a. Functionality	
b. Usability	
c. Reliability	

Total Score: _____

Action Taken: _____

Appendix C
SOFTWARE ACCEPTANCE CERTIFICATE

This certifies that the project entitled NUCATALOGUE: Web-Based Inventory Management System for NU LAGUNA Merchandise has been presented to us and hereby accepted with all its features, functionality, and performance. This also confirms that the project output fits the requirements of Asset Management Office as agreed during the conduct of the project.

This certification was issued to assure that the software project was free of errors during turnover and that the client accepts it to their satisfaction.

Signed:

Client Name: Christine P. Bonifacio

Position: Head Office Asset Management Office

Company: NU Laguna

Address: Km. 53 Pan-Philippine Hwy, Calamba 4029 Laguna

Project Title: NUCATALOGUE: Web-Based Inventory Management System for NU LAGUNA Merchandise

Proponents:

1. Cababasay, John Rick E.
2. Cayabyab, Christopher Jan L.
3. Ponti, Raven Arvey S.
4. Pregonero, Dave Andre S.
5. Ruedas, Hadji Aaron D.

Appendix D

CURRICULUM VITAE

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PERSONAL INFORMATION

Date of Birth:	August 28, 2001
Age:	21
Place of Birth:	Calamba City, Laguna
Height:	5’5
Religion:	Roman Catholic
Sex:	Male
Civil Status:	Single

EDUCATIONAL ATTAINMENT

Grade School:

Liceo de Calamba-Annex 2006-2014

High School

Immaculate Concepcion Catholic School 2014-2018

Senior High School

Saint Benilde International School 2018-2020

College

National University Laguna 2020-Present

Program: BS in Information Technology

Christopher Jan L. Cayabyab

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PERSONAL INFORMATION

Date of Birth:	July 25, 2001
Age:	21
Place of Birth:	Pasig City, Metro Manila
Height:	5’6
Religion:	Roman Catholic
Sex:	Male
Civil Status:	Single

EDUCATIONAL ATTAINMENT

Grade School:

Saint Isidore Children Center 2006-2014

High School

Active Community Contributors Institute 2014-2018

Senior High School

Asian Computer College - Senior High School 2018-2020

College

National University Laguna 2020-Present

Program: BS in Information Technology

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Date of Birth:	August 15, 2000
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Sex:	Male
Civil Status:	Single

EDUCATIONAL ATTAINMENT

Grade School:

Cabuyao Central Elementary School 2005-2013

High School

Arise and Shine Academy 2013-2017

Senior High School

Greenfields Integrated School of Laguna 2017-2019

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National University Laguna 2019-Present

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EDUCATIONAL ATTAINMENT

Grade School:

Casa Del Niño Montessori School 2006-2014

High School

Academia Institution De San Pedro 2014-2018

Senior High School

San Pedro College of Business Administration 2018-2020

College

National University Laguna 2020-Present

Program: BS in Information Technology

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Date of Birth:	August 4, 1999
Age:	23
Place of Birth:	Tananuan City, Batangas
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Sex:	Male
Civil Status:	Single

EDUCATIONAL ATTAINMENT

Grade School:

Marella Christian Institute	2007-2008
Tanauan Institute	2008-2012

High School

First Asia Institute of Technology and Humanities	2012-2013
Christian College of Tanauan	2013-2016

Senior High School

Lyceum of the Philippines University-Laguna	2016-2018

College

Lyceum of the Philippines University-Batangas	2018-2020
National University Laguna	2020-Present

Program: BS in Information Technology